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Multicast topology inference from measured end-to-end loss

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Abstract

The use of multicast inference on end-to-end measurement has recently been proposed as a means to infer network internal characteristics such as packet link loss rate and delay. In this paper, we propose three types of algorithm that use loss measurements to infer the underlying multicast topology: i) a grouping estimator that exploits the monotonicity of loss rates with increasing path length; ii) a maximum-likelihood (ML) estimator (MLE); and iii) a Bayesian estimator. We establish their consistency, compare their complexity and accuracy, and analyze the modes of failure and their asymptotic probabilities.

Real-World Impact & Application

Akamai & Global Content Delivery Networks: In the early commercial Internet, network operators could not see inside third-party ISP routers. Duffield et al. showed that sending synchronized multicast probe packets to end receivers allows mathematically inferring the internal router tree and detecting packet-dropping bottleneck links from the outside. This became the basis for modern CDN routing telemetry.